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Paper 1 of 1 — ENVIRONMENTAL ENGINEERING (PART 1) CHE 323

Level 300 • Second Semester • Handout

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
EFFURUN

DEPARTMENT OF CHEMICAL ENGINEERING

LECTURE NOTES

COURSE TITLE: ENVIRONMENTAL ENGINEERING (PART 1)

COURSE CODE: CHE 323

By

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Course Outline

1. Environmental pollution : Definition, natural and manmade pollution, the economics of pollution
2. Air pollution: gaseous and particulate pollutants and their sources, effects on whether and vegetation, materials and human health; legislation relating to air pollution
3. Method of control of gaseous emission; cyclones, inertia separators, electrostatic precipitator bag filters, wet washers etc. dispersal form chimney and flue stack
4. Water pollution: river pollution by industrial effluents, legislation and standards for effluent discharge, impurities in natural water and their effects

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1.0 Environmental pollution

1.1 Definition

Environmental pollution is defined as “the contamination of the physical and biological components of the earth/atmosphere system to such an extent that normal environmental processes are adversely affected. It is the addition of any substance (solid, liquid, or gas) or any form of energy (such as heat, sound, or radioactivity) to the environment at a rate faster than it can be dispersed, diluted, decomposed, recycled, or stored in some harmless form..

Environmental pollution is the major source of problems in human health and the sustainable development of society and economy in the world. The presence of environmental pollutants (i.e., heavy metals and persistent organic pollutants etc) in the environment poses grave danger to flora, fauna as well as humans

The main types of environmental pollution are:

□ Water Pollution. This type of pollution refers to the contamination of bodies of water including groundwater.

□ Air Pollution.

- ☐ Soil Pollution.
- ☐ Thermal Pollution.
- ☐ Radioactive Pollution.
- ☐ Noise Pollution.
- ☐ Light Pollution

These harmful substances introduced into the earth/atmosphere system are called pollutants. Pollutants can be natural or manmade

1.2 Manmade and Natural Pollution

The natural pollution are pollution resulting from natural phenomena e.g.: Volcanic eruptions and wild fires. Man-made pollution or pollution from anthropogenic sources are produced as a result of human activities.. Man-made pollution is generally a resultant of human actions such as consumption, waste disposal, industrial production, transportation and energy generation.

1.2,1 Manmade pollution

The burning of fossil fuels such as coal and gasoline is the single largest source of air pollutants. Burning these fuels causes smog, acid rain and greenhouse gas emissions. Burning fuels also increases some heavy metal contaminants and the amount of soot in the air. The harmful emissions spewed out by billions of passenger cars across the world and by highly pollutant power plants and factories have gradually degraded air quality, leading to respiratory problems but also to global warming and climate change. There are three types of manmade pollution:

- (a) Mobile sources – such as cars, buses, planes, trucks, and trains.
- (b) Stationary sources – such as power plants, oil refineries, industrial facilities, and factories.
- (c) Area sources – such as agricultural areas, cities, and wood burning fireplaces.

Manmade pollution can be mitigated by

- (i) Reducing the number of trips you take in your car.

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- (ii) Reducing or eliminating fireplace and wood stove use
- (iii) Avoiding burning leaves, trash, and other materials.
- (iv) Avoiding using gas-powered lawn and garden equipment

1.2,2 Natural Pollution

Natural pollution occurs naturally and won't cause excessive harm to our lives due to its regeneration ability. Listed below are some natural phenomena, though beneficial but can lead to natural pollution

- ☐ Defecation: Animals regularly put manure into the Earth through defecation. Although this can have beneficial effects on soil and generally does not cause significant problems in water, faeces can have disease-causing properties.
- ☐ Volcanic eruptions: Sulphur dioxide, nitrogen dioxide and nitrogen oxides can all be released into the atmosphere through the process of volcanic

eruption. This can lead to reduced air quality and enhance the onset of global warming.

□ Biological decay: Though the decay of organisms has led to the creation of fossil fuels and is a direct catalyst for the Industrial Revolution and the reason we are able to achieve such technological breakthroughs today, it can also lead to the emission of harmful gases into the atmosphere in a more direct manner. Nitrogen oxides and methane escape from decayed organic matter and compromise the environment.

□ Acidic rainwater: Rainwater is naturally acidic due to carbon dioxide present in the atmosphere. However, this acidity is exacerbated by our increased emissions of the contaminant, and gases expelled by volcanoes can make it even more damaging.

Of course, although these natural pollution processes have been taking place for millennia they do not pose a direct threat to the state of the environment. It is the contribution of humans that has had such a detrimental effect on our planet and could lead to grave complications for the flora and fauna which call it home.

1.3 The Economics of Pollution

Many countries including Nigeria are making concerted efforts to reduce pollution to the environment from both domestic and industrial activities especially the major source – fossil fuel. Despite the gradual reduction in emissions from fossil fuels, many important environmental issues remain. Along with the high levels of air and water pollution, other issues include hazardous waste disposal, destruction of wetlands and other wildlife habitats, as well as the impact of pollution on human health with economic repercussion.

Private markets, such as the sachet water industry, offer an efficient way to put buyers and sellers together and determine what quantity they produce, how they produce them and who gets them. The principle that voluntary exchange benefits both buyers and sellers is a fundamental building block of the economic way of thinking. However, what happens when a voluntary exchange affects a third party who is neither the buyer nor the seller? The effect of a market exchange on a third party who is outside or “external” to the exchange is called an externality. Because externalities that occur in market transactions affect other parties beyond those involved, they are sometimes called spillovers. Externalities can be negative or positive. If you hate Nigerian soap opera, then beaming it to you through a free

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channel every night would be a negative externality. However, if you love Nigerian soap opera, then what amounts to a series of free viewing would be a positive externality

1.3.1 Pollution as a Negative Externality

Pollution is a negative externality. Economists illustrate the social costs of production with a demand and supply diagram. The social costs include the private costs of

production that a company incurs and the external costs of pollution that pass on to society. Figure 1 shows the demand and supply for manufacturing refrigerators. The demand curve (D) shows the quantity demanded at each price. The supply curve (S_{private}) shows the quantity of refrigerators that all firms in the industry supply at each price assuming they are taking only their private costs into account and they are allowed to emit pollution at zero cost. The market equilibrium (E₀), where quantity supplied equals quantity demanded, is at a price of \$65,000 per refrigerator and a quantity of 45,000 refrigerators. Table 1 reflects this information in the first three columns. Taking Social Costs into Account - A Supply Shift

If the firm takes only its own costs of production into account, then its supply curve will be S_{private}, and the market equilibrium will occur at E₀. Accounting for additional external costs of \$10,000 for every unit produced, the firm's supply curve will be S_{social}. The new equilibrium will occur at E₁

Fig 1 Pollution cost of refrigeration production

Table 1: A Supply Shift Caused by Pollution Costs

Price	Quantity Supplied before	Quantity Supplied after
60	50,000	40,000
70	40,000	30,000
80	30,000	20,000
90	20,000	10,000

Considering Pollution Considering Pollution

Demanded

('000) Cost Cost

60 50,000 40,000 30,000

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Price	Quantity Supplied before	Quantity Supplied after
65	45,000	35,000
70	40,000	30,000
75	35,000	25,000
80	30,000	20,000
85	25,000	15,000
90	20,000	10,000

Considering Pollution Considering Pollution

Demanded

('000) Cost Cost

65 45,000 45,000 35,000

70 40,000 50,000 40,000

75 35,000 55,000 45,000

80 30,000 60,000 50,000

85 25,000 65,000 55,000

90 20,000 70,000 60,000

However, as a by-product of the metals, plastics, chemicals and energy that refrigerator manufacturers use, some pollution is created. Let's say that, if these pollutants were emitted into the air and water, they would create costs of \$10,000 per refrigerator produced. These costs might occur because of adverse effects on human health, property values, or wildlife habitat, reduction of recreation possibilities, or because of other negative impacts. In a market with no anti-pollution

restrictions, firms can dispose off, certain wastes absolutely free. Now imagine that firms which produce refrigerators must factor in these external costs of pollution—that is, the firms have to consider not only labor and material costs, but also the broader costs to society of harm to health and other costs caused by pollution. If the firm is required to pay \$10,000 for the additional external costs of pollution each time it produces a refrigerator, production becomes more costly and the entire supply curve shifts up by \$10,000.

As the Table illustrates, the firm will need to receive a price of \$70,000 per refrigerator and produce a quantity of 40,000—and the firm's new supply curve will be S_{social} . The new equilibrium will occur at E 1. Considering the additional external costs of pollution results in a higher price, a lower quantity of produced, and a lower quantity of pollution. Let us see how the principle works when noise pollution is involved

Table 2 shows the supply and demand conditions for a firm that uses trumpets for announcement on the streets when requested (Town crier). The measurement is based on the number of announcement made.

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Table 2: Supply and Demand Conditions for a Trumpet-Playing Firm

Quantity Supplied

Quantity without Quantity Supplied after

Price

Demande announcement the announcements the costs of the

=N=

d costs of the externality

externality

20 0 10 8

18 1 9 7

15 2.5 7.5 5.5

12 4 6 4

10 5 5 3

5 7.5 2.5 0.5

Step 1: Determine the negative externality in this situation. To do this, you must think about the situation and consider all parties that might be impacted. A negative externality might be the increase in noise pollution in the area where the announcement is made

Step 2: Identify the initial equilibrium price and quantity only taking private costs into account. Next, identify the new equilibrium taking into account social costs as well as private costs. Remember that equilibrium is where the quantity demanded is

equal to the quantity supplied.

Step 3: Look down the columns to where the quantity demanded (the second column) is equal to the “quantity supplied without announcement the costs of the externality” (the third column). Then refer to the first column of that row to determine the equilibrium price. In this case, the equilibrium price and quantity would be at a price of #10 and a quantity of 5 when we only take into account private costs.

Step 4: Identify the equilibrium price and quantity when we take into account the additional external costs. Look down the columns of quantity demanded (the second column) and the “quantity supplied after paying the costs of the externality” (the fourth column) then refer to the first column of that row to determine the equilibrium price. In this case, the equilibrium will be at a price of =N=12 and a quantity of 4

Step 5: Consider how taking into account the externality affects the equilibrium price and quantity. Do this by comparing the two equilibrium situations. If the firm is forced to pay its additional external costs, then street announcement with trumpet becomes more costly, and the supply curve will shift up.

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Remember that the supply curve is based on choices about production that firms make while looking at their marginal costs, while the demand curve is based on the benefits that individuals perceive while maximizing utility. If no externalities existed, private costs would be the same as the costs to society as a whole, and private benefits would be the same as the benefits to society as a whole. Thus, if no externalities existed, the interaction of demand and supply will coordinate social costs and benefits.

However, when the externality of pollution exists, the supply curve no longer represents all social costs. Because externalities represent a case where markets no longer consider all social costs, but only some of them, economists commonly refer to externalities as an example of market failure. When there is market failure, the private market fails to achieve efficient output, because either firms do not account for all costs incurred in the production of output and/or consumers do not account for all benefits obtained (a positive externality). In the case of pollution, at the market output, social costs of production exceed social benefits to consumers, and the market produces too much of the product.

We can see a general lesson here. If firms were required to pay the social costs of pollution, they would create less pollution but produce less of the product and charge a higher price.

1.3.2 Summary

□ Economic production can cause environmental damage. This tradeoff arises for all countries, whether high-income or low-income, and whether their economies are market-oriented or command-oriented.

- An externality occurs when an exchange between a buyer and seller has an impact on a third party who is not part of the exchange.
- An externality, which is sometimes also called a spillover, can have a negative or a positive impact on the third party.
- If those parties imposing a negative externality on others had to account for the broader social cost of their behavior, they would have an incentive to reduce the production of whatever is causing the negative externality.
- In the case of a positive externality, the third party obtains benefits from the exchange between a buyer and a seller, but they are not paying for these benefits. If this is the case, then markets would tend to under produce output because suppliers are not aware of the additional demand from others.
- If the parties generating benefits to others would somehow receive compensation for these external benefits, they would have an incentive to increase production of whatever is causing the positive externality.

2.0 Air Pollution

Air pollution is contamination of the indoor or outdoor environment by any chemical, physical or biological agent that modifies the natural characteristics. Air pollution

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may be considered as a mixture of solid particles and gases in the air. Car emissions, chemicals from factories, dust, and mold spores may be suspended as particles. Ozone gas is a major part of air pollution in cities. Air pollution may arise either from gaseous emission (Ozone (O_3), Nitrogen oxides (NOX), Carbon monoxide (CO) and Sulphur dioxide (SO_2) as well as particulate matters (PM₁₀, PM_{2.5}). Air pollutants may either be primary or secondary. Primary air pollutants are those directly emitted into the atmosphere, while secondary pollutants are those formed by the intermingling and reaction of primary pollutants e.g. smog

2.1 Gaseous Pollution

The gaseous criteria air pollutants of primary concern in urban settings include sulphur dioxide, nitrogen dioxide, and carbon monoxide; these are emitted directly into the air from fossil fuels such as fuel oil, gasoline, and natural gas that are burned in power plants, automobiles, and other combustion sources. There are certain percentage of gases present in the atmosphere. An increase or decrease in the composition of these gases is harmful to survival. This imbalance in the gaseous composition has resulted in an increase in earth's temperature, which results in a phenomenon known as global warming

2.2 Particulate Air Pollution

Particle pollution, also known as particulate matter or PM, is a general term for a mixture of solid and liquid droplets suspended in the air. The air we breathe indoors and outdoors always contain particle pollution. Some particles, such as dust, dirt, soot, or smoke, are large enough to be seen with the naked eye but others like

bacterial and airborne viral particles are usually not visible to the naked eye.

Particulate matter is separated into three main groupings: coarse particles (PM 10), fine particles (PM2.5), and ultrafine particles (PM0.1). These particle sizes, in general, differ in origin and health effects.

2.2.1 Coarse Particles (PM10)

The largest particle size contained in the air is PM10., also known as coarse particles, PM10 is defined as all particles with an aerodynamic diameter of 10 µm or smaller.

Some common examples of PM10 are:

- (i) Mold spores
- (ii) Bacteria
- (iii) Dust
- (iv) Smoke
- (v) Airborne viral particles

Because PM10 includes any particles smaller than 10 µm in diameter, PM 10 also contains fine particles (PM2.5) and ultrafine particles (PM0.1), though we usually think of a coarse range of particle sizes between 2.5 and 10 µm. The last few examples from the list above are also included in PM 2.5. Coarse particles, while not as dangerous as fine or ultrafine dust, can still pose a significant health threat. These particles can penetrate into our lungs and irritate our airways, nose, throat, and eyes. PM10 is usually created directly, with sources like construction work, road dust, or natural dust storms, rather than secondary, atmospheric sources.

2.2.2 Fine Particles (PM2.5)

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The next grouping we will discuss is fine particles, arguably the most well-known type of particulate matter. PM2.5 is a grouping of particles with an aerodynamic diameter of 2.5 µm or less, capable of penetrating deep into our lungs and even entering our bloodstream.

Fine particles can come from natural or human-made sources, like:

- Vehicle exhaust
- Wildfires
- Power plant emissions
- Other combustion activities

2.2.3 Ultrafine Particles (PM0.1)

The ultrafine particle (PM0.1) is generally smaller than fine dust, with an aerodynamic of 0.1 µm or smaller, and originates from similar sources as PM2.5.

2.3. Causes of Air Pollution

Following are the important causes of air pollution:

2.3.1 Burning of Fossil Fuels

Most of the air pollution takes place due to the burning of fossil fuels such as coal, oil, gasoline to produce energy for electricity or transportation. The release of

carbon monoxide in high level indicates how much fossil fuel is burned. This also emits other toxic pollutants in the air. Inhaling air induced with pollutants due to the burning of natural gas and fossil fuel reduces heart's ability to pump enough oxygen causing one to suffer respiratory illness.

2.3.2 Automobiles

The gases emitted from vehicles such as trucks, cars, buses, etc. pollute the environment. These are the major sources of greenhouse gases and also result in diseases among individuals.

2.3.3 Agricultural Activities

Ammonia is one of the most hazardous gases emitted during agricultural activities. The insecticides, pesticides and fertilizers emit harmful chemicals in the atmosphere and contaminate it.

2.3.4 Microbial Decaying Process

Manufacturing, chemical, and textiles industries release a large number of carbon monoxides, hydrocarbons, chemicals and organic compounds which contaminate our environment. Bacteria and fungi play a fundamental role in the biogeochemical cycles in nature. They are the key indicators of abnormal environmental conditions. Decaying of these microorganisms releases methane gas which is highly toxic. Breathing toxic gas like methane may lead to death.

2.3.5 Factories and Industries

Factories and industries are the main source of carbon monoxide, organic compounds, hydrocarbons and chemicals. These are released into the air, degrading its quality.

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2.3.6 Mining Activities

In the mining process, the minerals below the earth are extracted using large pieces of equipment. The dust and chemicals released during the process not only pollute the air, but also deteriorate the health of the workers and people living in the nearby areas.

2.3.7 Domestic Sources

The household cleaning products and paints contain toxic chemicals that are released in the air. The smell from the newly painted walls is the smell of the chemicals present in the paints. It not only pollutes the air but also affects breathing.

2.4 Effects of Air Pollution

The hazardous effects of air pollution on the environment include:

2.4.1 Diseases

Air pollution has resulted in several respiratory disorders and heart diseases among humans. The cases of lung cancer have increased in the last few decades. Children living near polluted areas are more prone to pneumonia and asthma. Many people die every year due to the direct or indirect effects of air pollution.

2.4.2 Global Warming

Due to the emission of greenhouse gases, there is an imbalance in the gaseous composition of the air. This has led to an increase in the temperature of the earth. This increase in earth's temperature is known as global warming. This has resulted in the melting of glaciers and an increase in sea levels. Many areas are submerged underwater.

2.4.3 Acid Rain

The burning of fossil fuels releases harmful gases such as nitrogen oxides and sulphur oxides in the air. The water droplets combine with these pollutants, become acidic and fall as acid rain which damages human, animal and plant life.

2.4.4 Ozone Layer Depletion

The release of chlorofluorocarbons, halogens, and hydro chlorofluorocarbons in the atmosphere is the major cause of depletion of the ozone layer. The depleting ozone layer does not prevent the harmful ultraviolet rays coming from the sun and causes skin diseases and eye problems among individuals.

2.4.5 Effect on Animals

The air pollutants suspend on the water bodies and affect the aquatic life. Pollution also compels the animals to leave their habitat and shift to a new place. This renders them stray and has also led to the extinction of a large number of animal species.

2.5 Air Pollution Control

The following are the measures that should be adopted to control air pollution:

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2.5.1 Minimise the use of Vehicles

People should avoid using vehicles for shorter distances. Rather, they should prefer public modes of transport to travel from one place to another, where such mode is available and efficient. This not only prevents pollution, but also conserves energy.

2.5.2 Energy Conservation

A large number of fossil fuels are burnt to generate electricity. Therefore, do not forget to switch off the electrical appliances when not in use. Thus, you can save the environment at the individual level. Use of energy-efficient devices such Compact fluorescent lamps (CFLs) also controls pollution to a greater level.

2.5.3 Use of Clean Energy Resources

The use of solar, wind and geothermal energies reduce air pollution at a larger level. Various countries, including India, have implemented the use of these resources as a step towards a cleaner environment. Other air pollution control measures include:

1. By minimizing and reducing the use of fire and fire products.
2. Since industrial emissions are one of the major causes of air pollution, the pollutants can be controlled or treated at the source itself to reduce its effects. For example, if the reactions of a certain raw material yield a pollutant, then the raw materials can be substituted with other less polluting materials.

3. Fuel substitution is another way of controlling air pollution. In many parts of the World, petrol and diesel are being replaced by CNG – Compressed Natural Gas fueled vehicles and electric powered vehicles. These are measures adopted in reducing or eliminating air pollution from car exhaust emission
4. Another way of controlling air pollution caused by industries is to modify and maintain existing pieces of equipment so that the emission of pollutants is minimized.
5. Sometimes controlling pollutants at the source is not possible. In that case, process control equipment can be used to control the pollution.
6. A very effective way of controlling air pollution is by diluting the air pollutants.
7. The best way of reducing the ill effects of air pollution is tree plantation. Plants and trees reduce a large number of pollutants in the air. Ideally, planting trees in areas of high pollution levels will be extremely effective.

2.6 Legislation Relating to Air Pollution

The U.S. Environmental Protection Agency (USEPA) which sets limits on certain air pollutants is the globally acceptable legislation applicable to environmental pollution issues. It also enforces federal laws on clean water and safe drinking water as well as regulations to limit the impact of businesses on the environment. The Federal Government of Nigeria in 1988 established the Federal Environmental Protection Agency (FEPA), based on the USEPA regulations. This regulating body is now domiciled in the Federal Ministry of Environment. The body's mandate is to protect, restore and preserve the ecosystem of the Federal Republic of Nigeria. Many states in Nigeria have their own environmental regulating bodies. For example, the Lagos

State Environmental Protection Agency Law, was enacted to establish the Lagos State Environmental Protection Agency (LASEPA). LASEPA's functions include monitoring and controlling the disposal of waste in Lagos State and advising the State Government on all environmental management policies. In order to key-in to the FEPA regulatory platform, many states in Nigeria had earmarked one day each month which they call environmental sanitation day. Environmental sanitation has been defined as: the control of all those factors in man's physical environment which exercise or may exercise a deleterious effect on his physical, mental or social well-being. In addition to ensuring the generally cleanliness of the environment, there are legislations that stipulate penalties against individuals and corporate bodies for unregulated discharge of all types of wastes (gaseous, solid and liquid) into the environment. However, enforcement of such regulations had become a major problem as a result of corruptive manipulations

3.0 Method of Control of Gaseous Emission

Gaseous emission can be controlled or managed by the following:

1. Using public transports
2. Turn off the lights when not in use.

3. Recycle and Reuse.
4. No to plastic bags.
5. Reduction of forest fires and smoking.
6. Use of fans instead of Air Conditioner.
7. Use filters for chimneys.
8. Avoid usage of cracker

The axiom for air pollution control is the 3-Rs, i.e REDUCE – REUSE – RECYCLE

3.1 Pollution Control Equipment

Common types of equipment for collecting fine particulates include cyclones, scrubbers, electrostatic precipitators, and baghouse filters. Once collected, particulates adhere to each other, forming agglomerates that can readily be removed from the equipment and disposed of, usually in a landfill

3.1.1 Cyclone Separator

A cyclone separator, also called a cyclonic dust collector, is a widely used air pollution control device that cleanses flue gases of particulate matter before such gases exit into the atmosphere. It is a method of collecting up to 99% of airborne waste in an easy-to-empty container beneath the cyclone. The key example of cyclone separation of gas-solid mixtures is the separation of powder particles from the outlet air from spray dryers and fluidized bed dryers, both to increase yield and to minimize air pollution

Cyclonic spray scrubbers are used to help in air pollution control and they use the features of a spray chamber and a dry cyclone to remove pollutants from gas streams. The inlet gas enters the chamber tangentially, swirls through in a corkscrew motion, and exits while liquid is sprayed inside the chamber. Large scale cyclones are used in sawmills to remove sawdust from extracted air. Cyclones are also used in oil refineries to separate oils and gases and in the cement industry as components of kiln preheaters. Cyclones are increasingly used in the household, as the core technology in bagless types of portable vacuum cleaners and central vacuum cleaners. Cyclones are also used in industrial and professional kitchen ventilation for

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separating the grease from the exhaust air in extraction hoods. Smaller cyclones are used to separate airborne particles for analysis. Some are small enough to be worn clipped to clothing, and are used to separate respirable particles for later analysis.

Figure 2 shows a typical cyclone

Fig. 2 Set-up of a typical cyclone

3.1.2 Electrostatic Precipitator

The electrostatic precipitator works by removing particles and smoke from a gas stream using an electrostatic charge. Introducing a water spray can reduce the electrical resistance of the dust particles, allowing them to accept the charge more easily. The advantage of the electrostatic precipitator is that it can be used for the

collection of both dry and wet impurities and has low operating costs. The collection efficiency of the device is high even for small particles. It can handle large gas volumes and heavy dust loads at low pressures. However, the electrostatic precipitator is not very flexible to changes in operating conditions once installed and it cannot control gaseous emissions

3.1.3 Inertial separators

Inertial separators separate dust from gas streams using a combination of forces, such as centrifugal, gravitational, and inertial. These forces move the dust to an area where the forces exerted by the gas stream are minimal. The separated dust is moved by gravity into a hopper, where it is temporarily stored. The three primary types of inertial separators are:

- (i) Settling chambers
- (ii) Baffle chambers
- (iii) Centrifugal collectors

Neither settling chambers nor baffle chambers are commonly used in the minerals processing industry. However, their principles of operation are often incorporated into the design of more efficient dust collector

3.1.4 Wet Scrubbers

Dust collectors that use liquid are known as wet scrubbers. In these systems, the scrubbing liquid (usually water) comes into contact with a gas stream containing dust particles. Greater contact of the gas and liquid streams yields higher dust removal efficiency.

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There is a large variety of wet scrubbers; however, all have one of three basic configurations:

1. Gas-humidification - The gas-humidification process agglomerates fine particles, increasing the bulk, making collection easier.
2. Gas-liquid contact - This is one of the most important factors affecting collection efficiency. The particle and droplet come into contact by four primary mechanisms:
 - (i) Inertial impaction - When water droplets placed in the path of a dust-laden gas stream, the stream separates and flows around them. Due to inertia, the larger dust particles will continue on in a straight path, hit the droplets, and become encapsulated.
 - (ii) Interception - Finer particles moving within a gas stream do not hit droplets directly but brush against and adhere to them
 - (iii) Diffusion - When liquid droplets are scattered among dust particles, the particles are deposited on the droplet surfaces by Brownian movement, or diffusion. This is the principal mechanism in the collection of submicrometre dust particles.
 - (iv) Condensation nucleation - If a gas passing through a scrubber is cooled

below the dewpoint, condensation of moisture occurs on the dust particles.

This increase in particle size makes collection easier.

3. Gas-liquid separation - Regardless of the contact mechanism used, as much liquid and dust as possible must be removed. Once contact is made, dust particulates and water droplets combine to form agglomerates. As the agglomerates grow larger, they settle into a collector.

The "cleaned" gases are normally passed through a mist eliminator (demister pads) to remove water droplets from the gas stream. The dirty water from the scrubber system is either cleaned and discharged or recycled to the scrubber. Dust is removed from the scrubber in a clarification unit or a drag chain tank. In both systems solid material settles on the bottom of the tank. A drag chain conveyor system removes the sludge and deposits into a dumpster or stockpile.

3.2 Chimney and Flue Gas Stack

3.2.1 Chimney

A chimney is an architectural ventilation structure made of masonry, clay or metal that isolates hot toxic exhaust gases or smoke produced by a boiler, stove, furnace, incinerator or fireplace from human living areas. Chimneys are typically vertical, or as near as possible to vertical, to ensure that the gases flow smoothly, drawing air into the combustion in what is known as the stack, or chimney effect. The space inside a chimney is called the flue. Chimneys are adjacent to large industrial refineries, fossil fuel combustion facilities or part of buildings, steam locomotives and ships. The term smokestack (colloquially, stack) is also used when referring to locomotive chimneys or ship chimneys, and the term funnel can also be used. The height of a chimney influences its ability to transfer flue gases to the external environment via stack effect. Additionally, the dispersion of pollutants at higher altitudes can reduce their impact on the immediate surroundings. The dispersion of pollutants over a greater area can reduce their concentrations and facilitate compliance with regulatory limits

3.2.2 Flue Gas Stack

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A flue-gas stack, also known as a smoke stack, chimney stack or simply as a stack, is a type of chimney, a vertical pipe, channel or similar structure through which combustion product gases called flue gases are exhausted to the outside air. Flue gases are produced when coal, oil, natural gas, wood or any other fuel is combusted in an industrial furnace, a power plant's steam-generating boiler, or other large combustion device. Flue gas is usually composed of carbon dioxide (CO₂) and water vapour as well as nitrogen and excess oxygen remaining from the intake combustion air. It also contains a small percentage of pollutants such as particulate matter, carbon monoxide, nitrogen oxides and sulphur oxides. The flue gas stacks are often quite tall, up to 400 metres (1300 feet) or more, so as to disperse the exhaust pollutants over a greater area and thereby reduce the concentration of the

pollutants to the levels required by governmental environmental policy and environmental regulation.

When the flue gases are exhausted from stoves, ovens, fireplaces, heating furnaces and boilers, or other small sources within residential abodes, restaurants, hotels, or other public buildings and small commercial enterprises, their flue gas stacks are referred to as chimneys.

4.0 Water Pollution

Water pollution may be defined as the contamination of streams, lakes, seas, underground water or oceans by substances, which are harmful for living beings. If the concentration of substances naturally present in water increases than acceptable level, the water is said to be polluted. Water pollution destroys important food sources and contaminates drinking water with chemicals that can cause immediate and long-term harm to human health. Water pollution also often severely damages aquatic ecosystems. Major sources of water pollution are:

- (i) Industrial waste
- (ii) Sewage and wastewater
- (iii) Mining activities
- (iv) Marine dumping
- (v) Accidental oil leakage
- (vi) Burning of fossil fuels
- (vii) Chemical fertilizers and pesticides
- (viii) Leakage from sewer lines.

There are three main categories of water pollution -: point-source, nonpoint-source and trans-boundary pollution

□ Point source pollution (PS) comes directly from a known source like an industrial or sewage outfall pipe. Point sources are typically associated with manufacturing processes. However, point sources also include discharges from water treatment plants and large animal feeding operations

□ Nonpoint source pollution (NPS) can be defined as pollution that comes from many miscellaneous or diffuse sources rather than from an identifiable, specific point. Nonpoint source pollution can originate from urban environments such as yards in Chemicals, waste products and soil that are carried by rain into streams or rivers become a part of NPS. Common examples are fertilizers,

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herbicides, pesticides, spilled motor oil and wastes from pets, wildlife and livestock.

□ Transboundary pollution is pollution that originates in one country but, by crossing the border through pathways of water or air, is able to cause damage to the environment in another country. Marine pollution is an excellent example of a transboundary pollution problem that involves many nation-states and unlimited point sources of pollution. Marine pollution can be the result of

on-shore industrial processes that use the ocean as a waste disposal site.

4.1 River Pollution by Industrial Effluents

Industries discharge chemical wastes comprising substances called effluents in rivers, lakes, streams etc. These waste effluents which are rich in decomposable organic matter, is the primary cause of organic pollution. Waste waters from crude oil production and processing, textile, brewery, food and beverages, paper, pulp and palm oil industries are believed to give a broad outline of industrial wastes as well as disposal problems. Factories sometimes turn waterways into open sewers by dumping oil, toxic chemicals and other harmful liquids called effluents into them. Industrial wastewaters entering a water body represent a heavy source of environmental pollution in Nigerian rivers. It affects both the water quality as well as the microbial and aquatic flora. Industrial effluents are characterized by their abnormal turbidity, conductivity, chemical oxygen demand (COD), total suspended solids (TSS), biological oxygen demand (BOD), and total hardness.

4.2 Effects of Industrial Effluents on the Environment

Industrial waste may pollute the nearby soil or adjacent water bodies, and can contaminate groundwater, lakes, streams, rivers or coastal waters. Industrial waste is often mixed into municipal waste, making accurate assessments difficult. Lakes, rivers and coastal areas are much more enjoyable and safe when they are pollution free. The negative impacts of industrial effluents to aquatic ecosystems and to humans, from harmful substances found in them are enormous. Some of these impacts can include death of aquatic life, algal blooms, habitat destruction from sedimentation and debris. Pollution of water resources by industrial effluents can interfere with swimming and fishing activities, make boating unsafe, affect wildlife and food resources and even contaminate water supplies.

4.3 Impurities in Natural Water and Their Effects

Biological impurities in water are caused by the presence of living organisms. These include algae, protozoa, pathogens, bacteria, viruses, microbes, and parasites along with their cysts (eggs) in contaminated water. The latter are simply called germs, microorganisms that can reproduce at alarming rates. The presence of impurities in natural drinking water can be done by the following:

□ Taste: The taste sense is moderately accurate and able to detect concentrations from a few tenths to several hundred ppm. ...

□ Color: Color is contributed primarily by organic material, although some metal ions may also tint water. ...

□ Odor.

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The most-deadly impurities found in water make the water extremely unsafe for use. Biological contaminants such as bacteria, protozoa, viruses, and parasites may lead to fatal waterborne diseases and health disorders. In addition to biological impurities,

natural water may also contain dissolved impurities such as calcium, sodium and magnesium chloride. Dissolved impurities imparts negatively on drinking water in the following ways:

- Makes the water hard in nature
- Changes chemical nature of the water
- Decreases the quality of the water
- Causes environmental degradation
- Causes loss of habitat for fresh water animals

4.4 Legislation and Standards for Effluent Discharge

Effluent standards, on the other hand, pertain directly to the quality of the treated wastewater discharged from a sewage treatment plant. The factors controlled under these standards usually include biochemical oxygen demand (BOD), suspended solids, acidity, and coliforms. Most legislation put in place is aimed at controlling effluent discharges into the environment including rivers. The control framework involves all stakeholders i.e. government agencies, scientists, industries and individuals. The common frameworks are as follows

Government agencies:

- Support research/education
- Enact laws that limit pollution levels
- Levy fines and penalties against polluters
- Coordinate state pollution control efforts
- Create environmental protection plans
- Provide mechanism to monitor pollution clean-up plans control programs

Industries:

- Support education programs, wildlife. preserves, etc.
- Establish quality control to limit pollution
- Develop recycling programs
- Research and use better production methods
- Monitor water quality of discharges
- Work with the general public to protect natural resources

Scientists:

- Identify sources and types of pollution
- Determine amount and concentration of pollution
- Study the effects of pollution
- Recommend safe pollution levels
- Study and design pollution control methods
- Develop pollution remediation and clean-up plans control programs
- Monitor effectiveness of clean-up efforts
- Research new treatment technologies

Individuals:

- Lobby for beneficial laws

- ☐ Educate public of pollution dangers
- ☐ Identify sources of pollution and notify authorities and public
- ☐ Encourage consumer conservation and recycling
- ☐ Volunteer to clean up polluted areas

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- ☐ Participate in citizen volunteer water quality monitoring programs
- ☐ Provide public information

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